

MONALI WASEKAR

Lead Front-End Developer | Angular, TypeScript, Nx

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SUMMARY

- Lead Front-End Developer for AspenTech's engineering-simulation platform — own Angular architecture across four applications spanning cloud web, embedded desktop (WebView2), and published component packages.
- Design backend-independent domain libraries distributed via Artifactory, so a single Angular UI runs against different calculation engines — in the browser and embedded inside the native HYSYS desktop product.
- Architect the shared library layer of a large Nx monorepo (17+ libraries) using RxJS, NgRx, Signals, GraphQL/Apollo, SignalR, and AG Grid Enterprise, with Jest, SonarQube, and Azure DevOps CI enforcing quality at scale.
- Author the engineering standards — Angular style, security (OWASP DOM-XSS), and testing conventions — adopted across frontend teams and encoded for AI coding agents to keep implementation consistent.
- Lead a small team through design and delivery, run front-end design reviews, and onboard engineers onto the Angular/Nx codebase.

CORE STACK

Angular 20/21, React 18/19 · TypeScript · Nx · RxJS · NgRx · Signals · Standalone Components · Angular Elements · Angular CDK · GraphQL/Apollo · SignalR
ngx-formly · ngx-translate · AG Grid Enterprise · ECharts · Shoelace / Web Components · Tailwind · WebView2 · React.JS · Jest / Spectator · SonarQube · Azure DevOps · Artifactory · GitHub Copilot

EXPERIENCE

Software Developer (Front-End) — AspenTech, MA

May 2022 – Present

NGE Front-End Simulator Platform | Angular, TypeScript, Nx

- Architected the shared library layer of the Nx monorepo (GraphQL, unit-of-measure, shared services, fluid-package flow, dynamic forms, physical properties), defining module boundaries and dependency rules so feature teams could compose apps without duplicating domain logic.
- Built a unit-of-measure (UOM) library driving conversion and display formatting for thermodynamic quantities across every screen, and a schema-driven dynamic forms library so backend-defined screens render without front-end code changes.
- Established the workspace testing and quality baseline — Jest/Spectator conventions, coverage gates, SonarQube static analysis, and Nx affected-graph builds in Azure DevOps — cutting CI time by rebuilding only impacted projects.
- Standardized the enterprise UI layer on the AspenTech design system and Shoelace web components, resolving multi-version design-system coexistence so apps could upgrade independently; reviewed PRs and set org-wide Angular conventions (standalone components, signals-based state, OnPush, RxJS).

Physical Properties Web-Based Application | Angular Elements, WebView2

- Designed a transport-agnostic service abstraction (base analysis/property services with GraphQL and HYSYS-host implementations) so a single UI library runs against either the NGE cloud backend or the

native HYSYS calculation engine — published to Artifactory as a backend-independent package.

- Implemented the parametric analysis engine in RxJS — concatMap/toArray pipelines sequencing temperature/pressure sweeps with per-point phase-probing, fallback logic, and back-pressure control against the host engine.
- Built an analysis result cache and configuration-persistence layer so users can switch tabs, reopen cases, and restore prior sweeps without re-running expensive thermodynamic calculations.
- Hardened the WebView2 bridge against untrusted input — origin/shape validation, safe JSON parsing, correlation-ID matching, timeout and error propagation — closing a DOM-XSS/message-spoofing risk class flagged in security review.
- Delivered property-set management on AG Grid Enterprise (custom cell renderers, action-button columns) and interactive ECharts visualizations; migrated the app to Angular Signals and standalone components, and automated the release path (version-stamped builds, npm publish to Artifactory) consumed by both the NGE app and the HYSYS desktop shell.

HYSYS Web Forms — Native Desktop UI Modernization | Angular 21, ngx-formly, WebView2

- Led the front-end for replacing legacy native HYSYS desktop dialogs with an embedded Angular application, rendered in-process via WebView2 inside the C++/C# simulator shell — enabling incremental UI modernization without rewriting the desktop product.
- Designed a schema-driven form runtime on ngx-formly: the C# engine emits a field model and the Angular layer generates the full screen at runtime, so new engine dialogs ship with zero front-end changes.
- Authored a library of 30+ custom Formly field types covering the simulator's UI vocabulary — tab controls, group boxes, matrix widgets, plot fields, attachment lists/trees, unit-aware enumerations, and embedded view frames.
- Built a field reconciler that diff-merges incoming engine schemas against live form state, preserving focus, navigation, and scroll position across engine-driven rebuilds; implemented the host bridge service (subscribe/postMessage/postMessageAndWait) with debounced updates to keep engine and UI in sync.
- Applied OWASP DOM-XSS guidance to all engine-originated payloads (schema-validated parsing, no raw HTML binding) and codified the rules as repo-level security and Angular style conventions for the team.

AIMB — Inferential Model Builder | Angular, Nx

- Built core AIMB screens from the ground up — Data Set management, Build Model configuration (variable mapping, constraints), data-cleaning workflows (filtering, plots, steady-state validation), and Analyze/Deploy pages.
- Delivered end-to-end project import/export and owned search/filtering across the app; rolled out features behind toggles for staged QAV/Production releases, with i18n coverage (ngx-translate) and unit tests (Jest/Spectator) alongside feature work.

PROJECTS (FREELANCE / COMMUNITY)

gënie — Household Inventory App | React, TypeScript, React Native (Expo)

- Designing and building a shared household inventory and shopping-list app that syncs item stock levels in real time across family members, so anyone can mark an item low or out and the whole household sees it instantly.
- Architecting a single-codebase strategy (Expo + React Native Web) so the same TypeScript component and business-logic layer ships to both web and native mobile from one repository.
- Building the real-time data layer for multi-user household sync, live shopping-mode list updates, and in-app household broadcast messaging; planning OCR-based receipt scanning and location-based restock

alerts as follow-on features.

- Designed the product concept end-to-end, including UI/visual design direction and a public landing page, prior to development.

Gift Card Wallet | React, Redux, Node.js

- Built a digital gift-card platform letting users buy, send, and manage gift cards from major retailers (Amazon, iTunes, Starbucks, Target) as a single-page app using React, Redux, and ES6.
- Built reusable React components and a React-Router-based SPA architecture; used Node.js/Express and MongoDB to build RESTful APIs for the client.

Camping Website | React, React Native

- Built a campsite discovery, review, and registration site (React), with responsive forms, tooltips, and drag-and-drop widgets, extended to mobile with React Native.

EDUCATION

NuCamp Bootcamp — Full Stack Web & Mobile Development	2021
Master's in Computer Management — Pune University	2013
Bachelor's in Computer Science — Nagpur University	2010